



COLLEGE
OF
MEDICINE

Infertility

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:OBJECTIVE

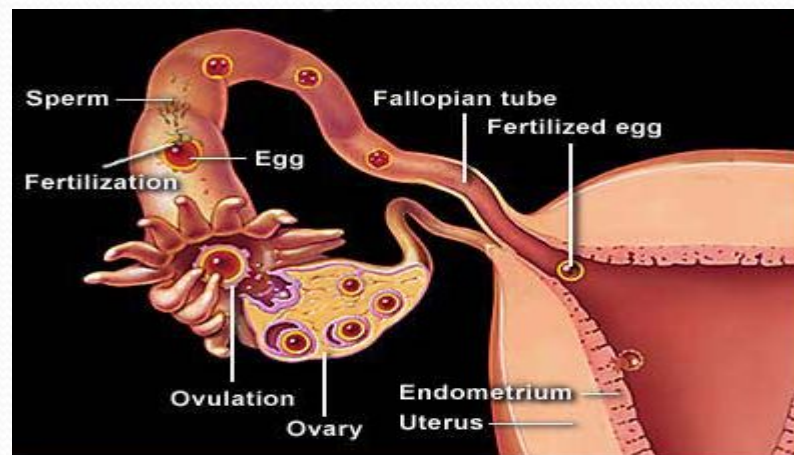
- • Understand the definition and causes of subfertility.
- • Describe the concept of ovarian reserve.
- • Understand the history, examination and investigations relevant to subfertility.

:Introduction

- **Fifteen per cent** of couples who want a baby experience an unwanted delay in conception.
- Although there has been no change in the prevalence of fertility problems, more couples seek help than did previously.
- The causes of fertility problems include **disorders of ovulation, sperm and the Fallopian tube**, although no identifiable cause is found in a third of couples trying for a baby

Requirements for Conception

- normally developed reproductive tract in both the male and female partner
- normal functioning of an intact hypothalamic-pituitary-gonadal axis supports *gametogenesis* (**the formation of sperm and ova**).
- timing of intercourse
- **Unblocked tubes** that allow sperm to reach the egg
- The **sperms** ability to penetrate and fertilize the egg
- **Implantation** of the embryo into the hormone-prepared endometrium
- Finally a **healthy pregnancy**



Introduction

- **Ferti'lity treatment may be**
- **medical,**
- **surgical or**
- **involve assisted conception whereby the egg and sperm are brought into close proximity to facilitate fertilization.**



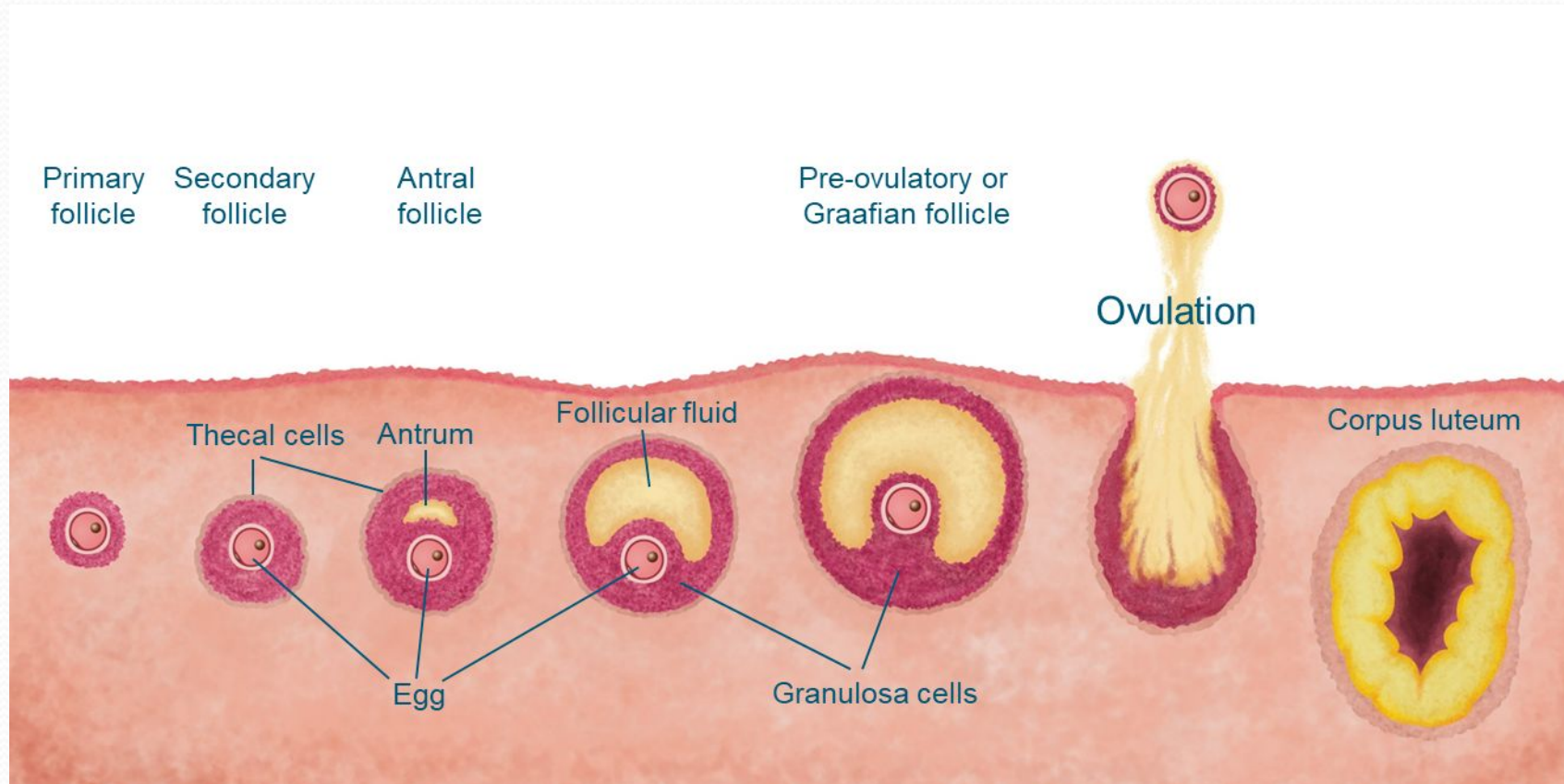
Introduction:

- Infertility causes great distress to many couples, causing increasing numbers of them to seek specialist fertility care.
- Most of those presenting with childlessness have reduced fertility, rather than absolute sterility, and many are likely to conceive spontaneously.

BACKGROUND INFORMATION

- At puberty there are **300,000** primordial follicles
- Dominant follicle produces oestradiol which leads to LH surge
- Ovulation occurs **24-36** hours later
- The fertilization life span of the ovum is **24-36** hours
- The receptivity of the endometrium is days **16-19** of a **28** day cycle

:The ovarian follicle





:Natural conception

Women with a normal menstrual cycle of **28 days** ,
.ovulation occurs around day **14**

The average survival time of the oocyte is around **24**
hours , while after ejaculation sperm may survive
. **days** in the female reproductive tract **7 – 2**

: Epidemiology

Up to **90%** of couples will have conceived for after regular unprotected intercourse .three years

- In the general population, conception is expected to occur in **84%** of women within 12 months and in **92% within 24 month.**



Definition of Infertility :

is defined as the inability to conceive after one to two . years of unprotected intercourse

Data suggest that 10–15% of couples experience infertility , Half of them (8%) will subsequently conceive without the need for specialist advice and treatment.

◎ **Of the remaining 8% who require input from fertility clinics, Half (4%) comprise couples with primary infertility (no history of a previous pregnancy),**

while the other half have secondary infertility (difficulties in conceiving after an initial pregnancy).

Infertility is commonly divided into five major :categories

● **Diagnostic categories in infertility:**

● Anovulation	20 %
● Male	25 %
● Tubal	15 %
● Endometriosis	10 %
% Unexplained	30

**The likelihood of spontaneous live birth in
: infertile couples is strongly influenced by**

- **1. female age,**
- **2. duration of infertility,**
- **3. previous pregnancies,**
- **4. and cause of infertility .**



Factors adversely affecting conception rates

Combined factors	Male factors	Female factors
Duration of infertility (years 2 <) No previous conception	Low numbers ,of motile healthy sperm Drug intake	Age (> 37 years) Menstrual FSH level (> 10 u/L)

Causes of female infertility : include

1. **Ovulatory disorders secondary to ovarian dysfunction .**
2. **Tubal disease and blockage .**
3. **Endometrial factors.**
- 4- **Hypothalamic- pituitary – ovarian (HPO) axis dysfunction.**

:Ovulatory disorders

- Absence of ovulation (**anovulation**) or infrequent ovulation (**oligo-ovulation**) is seen in a **fifth** of all women presenting with infertility.
- Abnormalities of gonadotrophin releasing hormone (**GnRH**) agonist secretion are associated with **very low levels of oestradiol, follicle stimulating hormone (FSH) and luteinizing hormone(LH).**

Ovulatory factor

- **Kallman's Syndrome** is a congenital cause of anovulation characterized by isolated gonadotrophin deficiency and anosmia.
- Acquired causes include pituitary tumours, pituitary necrosis (**Sheehan's syndrome**), stress and excessive weight loss or exercise.
- Clinical examination of the visual fields and imaging of The pituitary fossa are indicated when a space occupying pituitary lesion is suspected

:Ovulatory factor

- Other causes of an ovulation occurs in the majority of women with normogonadotrophic anovulation Have polycystic ovary syndrome(PCOS).
- **Diagnosis of PCOS includes the presence of two out of the three listed below:**
 - 1. Oligo- and/or anovulation.**
 - 2. Clinical and/or biochemical signs of hyperandrogenism.**
 - 3. Polycystic ovaries.**

Ultrasound of a polycystic ovary showing dense stroma : and peripheral cysts



- Other causes of unovulation is amenorrhoea with elevated serum **FSH** and **low** or undetectable oestrogen levels signify ovarian failure. Known causes include :
- **Turners Syndrome (XO),**
- **gonadal dysgenesis,**
- **autoimmune disorders,**
- **irradiation or chemotherapy,** in many cases the cause is unknown.

HYPERPROLACTINAEMIA

Increased levels of prolactin interfere with normal pulsatile secretion of GnRH, resulting in **anovulation, amenorrhoea and occasionally galactorrhoea associated with low FSH and oestradiol levels.**

Hyperprolactinaemia is a feature of prolactin producing pituitary adenomas or tumours blocking inhibitory control of the hypothalamus.

Other causes include primary hypothyroidism, chronic renal failure, and drugs such as the combined oral pill, dopamine depleting agents (**reserpine, methyldopa**) and dopamine receptor inhibiting agents (**metoclopramide**).

Tubal factor infertility

Tubal disease accounts for **15–20%** of cases of primary infertility and approximately **40%** of secondary infertility.

It represents the aftermath of pelvic infection or surgery resulting in tissue damage, scarring and adhesion formation.

Tubal factor infertility

- This can affect tubal function and result in either partial or total tubal occlusion.
- As the distal portion of the tube is commonly affected, fluid can accumulate within the tubes causing a hydrosalpinx.

Functional competence of the fallopian tubes implies not just patency but also the integrity of the mucosal lining or the endosalpinx.

Tubal factor infertility

- any damage to the fallopian tubes tends to be irreversible
- correction can be difficult.
- **Due to current limitations in investigating tubal function it is only possible to assess the macroscopic appearance and patency of the fallopian tubes.**

Tubal factor infertility

- The principle cause of tubal disease is pelvic inflammatory disease (PID) which may occur spontaneously or as a complication of miscarriage, puerperium, intrauterine instrumentation and pelvic surgery.

A single episode of PID carries up to 10% risk of future tubal factor infertility

Tubal factor infertility

- The risk is aggravated by further infections due to
- *Chlamydia trachomatis* or *Neisseria gonorrhoeae*.
- Chlamydia is now the most common sexually transmitted disease (STD) and responsible for at least 50% of identifiable cases of PID.
- Lower abdominal surgery is a risk factor for tubal infertility.

Tubal factor infertility

- Most abdominal and pelvic surgery causes adhesions.
- Gynaecological surgery, appendicectomy, bowel resection and urological operations are all thought to increase the risk of subsequent tubal disease.
- a number of studies reported an increased risk of PID in women who used IUCDs as compared to non-users.

Tubal factor infertility

- Congenital abnormalities are **uncommon** causes of tubal pathology and are associated with developmental anomalies of the urinary system.
- **Endometriosis, cornual fibroids or polyps** can cause cornual block or tubal distortion.

Endometriosis

- Endometriosis is characterized by the presence of uterine endometrial tissue outside the cavity of the uterus.
- The common sites are the pelvic peritoneum, ovaries and rectovaginal septum.
- **The prevalence of pelvic endometriosis in women with infertility has been shown to be 21%.**

Unexplained Infertility



Unexplained infertility

Unexplained infertility is diagnosed where routine investigations including **semen analyses, tubal evaluation and tests of ovulation yield normal results.**

- the reported prevalence of unexplained infertility, report incidences of **20–30%.**
- Failure of routine tests to detect any obvious contributory factors has led clinicians to speculate about factors contributing to a diagnosis of unexplained infertility.

Male factor infertility

- The male partner is directly responsible for **25%** of cases of infertility and is thought to play a contributory role in **another 25%**.
- **Male factor infertility** implies a lack of sufficient numbers of competent sperm, resulting in failure to fertilize the normal ovum

normal semen parameters(WHO)

Parameter

Normal value

- -Volume 2.0 ml or more
- -PH 7.2 – 7.8
- - Sperm concentration $20 \times 10^6/\text{ml}$ or more
- -Motility 50% or more with progressive motility (Grade a or b)*
- -Morphology 15–30%†
- -Viability 75% or more live
- -White blood cells Fewer than $1 \times 10^6/\text{ml}$
- * Grade a: rapid progressive motility; Grade b: slow or sluggish motility.

Nomenclature for some semen variables

- **Normozoospermia:** normal ejaculate as defined by the reference value.
- • **Oligozoospermia:** sperm concentration less than the reference value .
- • **Asthenozoospermia:** less than the reference value for motility.
- • **Teratoloospermia:** less than the reference value for morphology.
- • **Azoospermia:** no spermatozoa in the ejaculate.
- • **Aspermia:** no ejaculate.

:Causes of male infertility

- 1-No demonstrable cause
- 2-Varicocoele
- 3-Idiopathic oligozoospermia .
- 4-Accessory gland infection .
- 5-Idiopathic teratozoospermia .
- 6-Idiopathic asthenozoospermia .

Causes of male infertility

- 7-Suspected immunological infertility .
- 8-Systemic diseases
- 9-Obstructive azoospermia .
- 10-Ejaculatory inadequacy
- 11-Hyperprolactinaemia .
- 12-Iatrogenic causes.
- 13-Pituitary lesions ,Gonadotrophin deficiency



Management of infertility:

Management of infertility:

- Couples should be seen when a fertility problem is perceived to exist.
- This first consultation can be in primary care and does not necessarily require referral to a specialist clinic.

Management of infertility

- Exclusion of any obvious medical factors,
- explanation about normal patterns of conception and advice about lifestyle measures may be sufficient in many cases.
- Referral to a fertility clinic should take into account **the age of the female partner and duration of infertility**.

Management of infertility

- In the absence of any known reproductive pathology, couples who have been trying for **1–2 years** should be investigated and seen in a dedicated fertility clinic.
- **Earlier intervention is indicated in the presence of specific high-risk factors in either partner.**

Management of infertility

- In the male, this could be a history of **azoospermia, testicular surgery, vasectomy or coital failure.**

Reasons for early referral in a **woman** include **oligoamenorrhoea, known endocrine** conditions affecting ovulation

Management of infertility

- History of tubal disease, endometriosis or salpingectomy.
- Accessing fertility care is a joint decision for couples who should be encouraged to attend together.

Proposed investigations and treatment should be ,information explained by adequate verbal and written **consideration should be given to the social and .psychological needs of couples**

:History



History

- A detailed history should be elicited from both partners. This should include questions about the
,duration of infertility
,general health
past medical and
surgical history and
specific questions about sexual history

History:

Male:

- Evidence of previous fertility with past partners
- Previous investigations or treatment for infertility
- Medical Sexually transmitted diseases
 - .Mumps orchiditis
- Testicular maldescent
- Chronic disease or medication
- Drug/alcohol abuse
- Recurrent urinary tract infection (UTI)

History in male

- **Surgical history:**

- Testicular torsion
- Orchidopexy
- Testicular injury
- Vasectomy and vasectomy reversal
- Occupational Exposure to toxins
- Sexual Decreased libido
- Impotence

:Female (History)

- Fertility in previous relationships
- Time to previous conceptions
- Previous fertility investigations or treatments
- Length and type of previous contraceptive use
- Menstrual history Cyclicity
- Amenorrhoea
- Dysmenorrhoea
- Heavy menstrual bleeding
- Intermenstrual bleeding

: History (female)

- **Obstetric history**: Previous pregnancy Miscarriage, ectopic pregnancy.
- Medical history :Chronic illnesses (diabetes, hypertension ,renal disease)
- Known endocrine disorders, e.g. hypothyroidism, PCOS
- Previous STD's, e.g. Chlamydia
- Known endometriosis ,Galactorrhoea

: History (female)

- Cervical smear history
 - _Surgical history :Tubal surgery including salpingectomy and salpingostomy
- Ovarian surgery
- Pelvic surgery for endometriosis
- Previous laparoscopy
- Appendicectomy
- Sexual history Coital frequency and timing

Examination of the infertile couple

- Female

- General examination

Height, weight, body

- mass index (BMI)

- Blood pressure Blood pressure

- Fat and hair distribution

- Acne and galactorrhoea

Male

weight, BMI

Female

- **Local examination**

- *Abdominal examination*

Scars

- *Pelvic Genitalia*

- Inspection of external genitalia

Male

Groin

Hernia

Abdominal masses

:Examination

- **Speculum examination:**
 - vaginal assessment –
 - vaginal septa, infections
 - Cervix – ectopy, polyps
- Bimanual palpation of uterus: size, shape, position, mobility.
- Presence of adnexal masses and tenderness.

Initial investigations

MALE:

- **Semen analysis** remains the most commonly performed investigation in the male.
- To adjust for fluctuations in semen parameters, a minimum of two samples **4 weeks** apart should be analysed. Samples should be collected after a period of **2–7 days of abstinence**.

FEMALE :Investigation

- -Anormal menstrual cycle is suggestive of ovulation.
- Confirmation of ovulation is usually obtained by means of **a mid-luteal serum progesterone level in excess of 30 nmol/l** 7 days before the onset of menstruation (day 21 of a 28 day cycle).

:Investigation

- In addition to tests of ovulation, a rubella screen should be performed on each woman.
- There is little evidence that routine use of temperature charts and LH detection kits improves clinical outcome.
- **There is no justification for routine assessment of FSH, LH, prolactin and thyroid function in ovulatory women.**

:Investigation

- **Mid cycle ultrasound** in day **12-13** of cycle for size of graffian follicle which should be between (**16 – 22 mm**), indicate mature ova, good ovulatory factor.

TUBAL PATENCY TESTING

- Once preliminary investigations suggest that a woman is ovulating and semen parameters are satisfactory,
- the next step should be assessment of tubal status.

- **Laparoscopy and chromotubation (lap and dye)** is the investigation of choice as it is able to demonstrate tubal patency well as assess the pelvis for the presence of endometriosis and adhesions.
- **Hysterosalpingogram (HSG):** which involves a pelvic X-ray following the injection of a **radio opaque iodine-based** dye through the cervix is less invasive and can be helpful in cases where laparoscopy is contraindicated/hazardous or in women at low risk of pelvic pathology

:CONCLUSION

- Infertility have psychological and social problem .
- Female and male infertility near equal.
- Infertility problem can be treated.



Thank You

!for attention

